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***B.Tech. Degree II Semester Regular and Supplementary/
I Semester Supplementary Examination April 2018***

**CE/ME GE 15-1203 A ENGINEERING GRAPHICS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

(5 × 12 = 60)

- I. Draw an ellipse having major axis of 75 mm and minor axis of 50 mm by intersecting arc method. Also draw a tangent and normal to the curve from any point 'p' on the curve.

OR

- II. Construct a vernier scale, $RF = \frac{1}{40}$ to show centimeters and long enough to measure up to 6 m. Indicate a length of 4.66 m and 2.37 m on this scale.
- III. The end 'Q' of a line PQ is in HP, length of its top view is 64 mm and it makes an angle of 30° with XY line. The end P is in VP and 40 mm above the HP. Draw its projections find true length, θ and ϕ .

OR

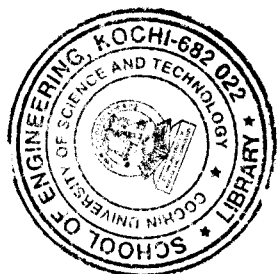
- IV. Side AB of a regular pentagonal lamina ABCDE, is in the VP with the end A 24 mm above HP. The end 'D' is 60 mm in front of the VP. Draw its projections if the side AB is inclined at 30° to the HP. The length of the side of the lamina is 50 mm.
- V. A frustum of a cone of diameter of base 60 mm, diameter of top surface 30 mm and axis 45 mm long is lying on HP on one of its generators such that the plane containing the axis and the generator makes an angle of 45° to the VP. Draw its front and top views.

OR

- VI. A right regular pentagonal pyramid side of base 30 mm and height 65 mm lies on one of its triangular face on HP, such that its axis is parallel to VP. A section plane, perpendicular to VP and parallel to HP, cuts the axis of pyramid at a point 7 mm from its base. Draw its front view and sectional top view.
- VII. A square prism, edge of base 30 mm and height 60 mm, resting on its base on HP, is completely penetrated by another square prism of 20 mm base edge such that the axis of the penetrating prism is perpendicular to end 5 mm in front of the axis of the vertical prism. The rectangular faces of the two prisms are equally inclined to the VP. Draw the projections of the solids showing line of intersection. Assume any suitable length for penetrating prism.

OR

(P.T.O.)



- VIII. A right rectangular pyramid base of side 40 mm and height 62 mm resting on its base on HP with one of its base side parallel to VP. A section plane perpendicular to VP and inclined at 30° to HP cuts the pyramid bisecting the axis. Develop the lateral surface of the truncated pyramid.
- IX. A regular triangular prism 40 mm base side and 80 mm long lying on one of its rectangular face on the ground and axis perpendicular to picture plane. One of the triangular faces is parallel to and 15 mm behind the picture plane. The observer is viewing the object from a point 50 mm above the ground and 30 mm in front of the picture plane. The centre plane is 90 mm towards the left of the axis of the prism. Draw the perspective view of the prism.
- OR**
- X. Draw the isometric projection of a sphere of 60 mm diameter resting centrally on top of a square slab of 60 mm $\times$ 60 mm $\times$ 20 mm.
